

UNIT 6

OUR EARTH, SUN, AND MOON

Learning objectives:

- Describe how the Sun's position in the sky changes throughout the day.
- Explain how shadows change with the Sun's position.
- Understand how Earth's rotation causes day and night.
- Describe Earth's rotation and how it affects the Sun's movement in the sky.
- Explain what an axis is and how Earth's tilt affects day length.
- Describe why seasons change due to Earth's revolution around the Sun.
- Identify how the Sun's path changes with seasons.
- Explain how seasons affect different regions in Pakistan.
- List and describe the phases of the Moon.
- Explain why the Moon's shape seems to change.
- Understand the importance of the lunar cycle in the Islamic calendar.

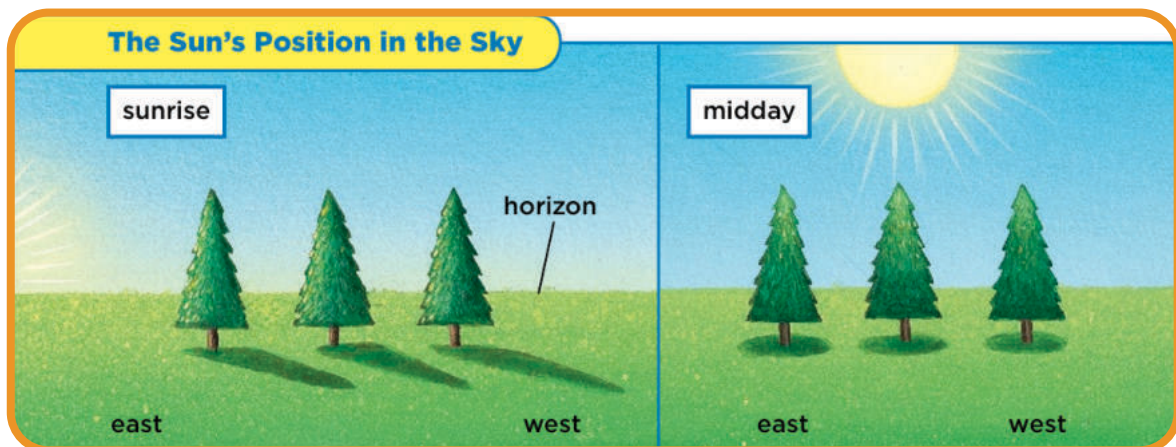


**How do the Earth and the Moon
move through space?**

6.1 DAY AND NIGHT

- **How does the Sun's position in the sky seem to change?**

The Sun seems to rise in the east each morning. It appears higher and higher in the sky. At midday, it is high overhead. Then it appears to sink in the west. The Sun appears to set in the west at the end of the day.



Shadows change with the position of the Sun in the sky.

- **At midday**, sunlight comes from high above, making shadows **very short**.
- **At sunrise**, light comes in at a low angle from the **east**, and shadows are **long, stretching west-northwest**.
- **At sunset**, the Sun is in the **west**, and shadows are **long, stretching east-northeast**.

Shadows grow longer →
as the Sun sets.



- **What causes night and day?**

Our planet Earth is always moving. It **rotates**, or turns. It takes **24 hours**, or **one full day**, to make one turn. As the Earth rotates, one side faces the **Sun**. It is **daytime** on that side. The other side faces away from the **Sun**. It is **night time** on that side.

Day and Night in Pakistan

- During the day, Pakistan faces the Sun, making it daytime.
- During the night, Pakistan is turned away from the Sun, making it night time.

Earth's Rotation and the Sun's Movement

As the Earth turns in space, the Sun appears to move across the sky. The Earth rotates from west to east, which makes the Sun seem to move from east to west.

In Pakistan, the Sun rises in the east every morning and sets in the west every evening. This happens because of Earth's rotation.



↑ The Sun rises in the east behind the mountains in the north of Pakistan. It will rise again in 24 hours.



- **What is an Axis?**

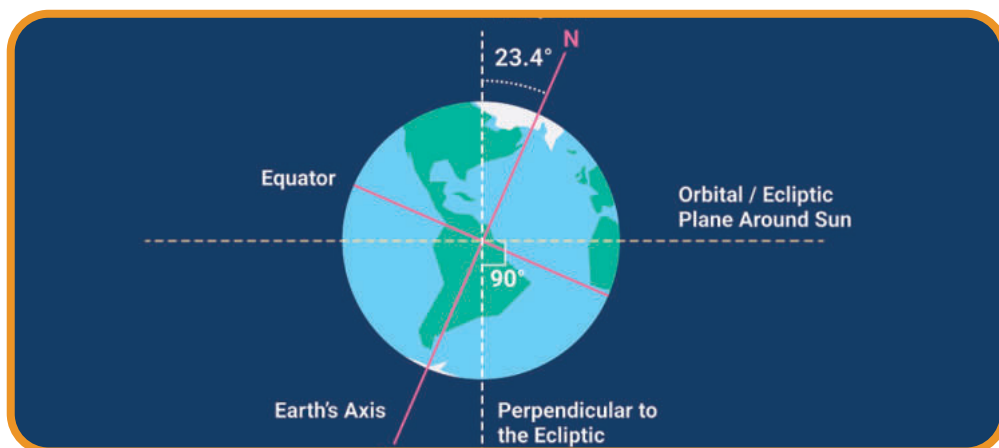
A ball can spin on your fingertip. It spins around an **axis**. An axis is a line through the center of a spinning object. The ball's axis starts at your fingertip and extends to the top of the ball.



↑ A ball spins around its axis.

Similarly, the Earth also has an axis. It rotates around this invisible line. Earth's axis is tilted, which affects how sunlight reaches different parts of the planet.

The North Pole is at the top of the Earth's axis, and the South Pole is at the bottom.



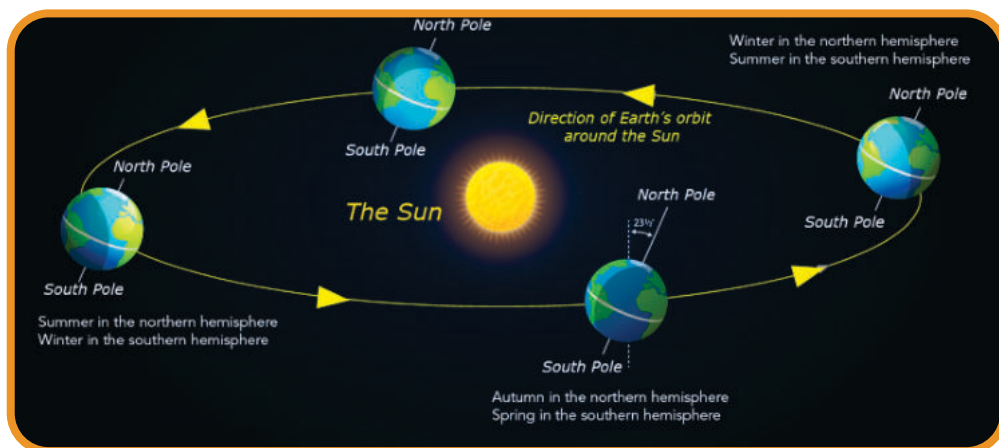
↑ In Pakistan, the Earth's tilt affects the length of days and nights throughout the year



• Why Do Seasons Change?

The Earth does not just rotate in space; it also revolves around the Sun. This means it follows a circular path called an orbit.

- The Earth takes 365 days to complete one orbit around the Sun. That equals one year.
- As the Earth moves, different areas receive varying amounts of sunlight, which causes seasonal changes.



↑ Revolvment of the Earth around the Sun.

In Pakistan:

- Summer (**May – August**) happens when Pakistan tilts towards the Sun, receiving more direct sunlight.
- Winter (**November – February**) occurs when Pakistan tilts away from the Sun, getting less sunlight.
- Spring (**March – April**) and Autumn (**September – October**) are transition seasons with moderate temperatures.



How Does the Sun's Path Change?

The Sun appears at different positions in the sky throughout the year.

- In **summer**, the Sun is **higher** in the sky, resulting in longer days.
- In **winter**, the Sun is **lower**, causing shorter days and longer nights.

How Do Seasons Affect Pakistan?

- In **summer**, Pakistan experiences **hot temperatures**, especially in **Sindh and Punjab**.
- In **winter**, the **northern areas** receive snowfall, while the **plains** have cold, dry weather.
- **Spring** brings moderate temperatures and blooming flowers.
- **Autumn** is a transition to winter, with falling leaves and cooler temperatures.

6.1 THE MOON

• What are the phases of the Moon?

Some nights, the **Moon appears as a full circle**. On other nights, only part of it is visible. The different **shapes** we see are called the **phases** of the Moon.





The **Moon has 8 main phases**, which change as it orbits Earth.

In Pakistan, the Moon's phases are important for Islamic events, such as determining the start of Ramadan and Eid.

Phases of the Moon

- **New Moon** – The Moon is not visible.
- **Crescent Moon** – A small curved part of the Moon is visible.
- **First Quarter Moon** – Half of the Moon is visible.
- **Waxing Gibbous Moon** – More than half is visible.
- **Full Moon** – The entire Moon is visible.
- **Waning Gibbous Moon** – The lit portion starts to shrink.
- **Last Quarter Moon** – Half of the Moon is visible again.
- **Waning Crescent Moon** – A small curved part is visible before the cycle restarts.

- **Why does the Moon's shape seem to change?**

The Moon is shaped like a ball and revolves around Earth. This movement makes the shape **appear to change**.

- **Half of the Moon is always lit by the Sun.** The other half remains dark.



- The **Moon takes about four weeks** (29.5 days) to complete one orbit.
- This cycle is called the **lunar cycle**.

In Pakistan, the Islamic calendar follows the lunar cycle.

A new Islamic month starts when the **new Moon** is sighted.



↑ Full Moon

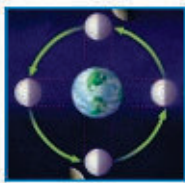


Draw a line between the word and its definition.



1. axis

to spin or turn



2. lunar cycle

the path an object takes as it moves around another object



3. orbit

any shape of the Moon that we see



4. phase

a line through the middle of a spinning thing



5. revolve

to move around another object in a circular path



6. rotate

the time it takes the Moon to go through all its phases



Test Your Understanding

Q1. Why does the Sun appear to move across the sky during the day?

Ans: _____

Q2. What causes day and night on the Earth?

Ans: _____

Q3. How do shadows change throughout the day?

Ans: _____

Q4. What is the Earth's axis, and how does it affect day length?

Ans: _____

Q5. Why do seasons change on the Earth?

Ans: _____

Q6. How does the Earth's tilt affect the weather in Pakistan?

Ans: _____



Q7. What is the lunar cycle, and how long does it take to complete?

Ans: _____

Q8. Why does the shape of the Moon seem to change?

Ans: _____

Q9. How is the lunar cycle important in the Islamic calendar?

Ans: _____

Q10. What happens to the Sun's position in the sky during summer and winter?

Ans: _____



Fill in the blanks.

1. The Earth rotates around an _____.
2. The Earth rotates from _____ to _____.
3. One day on the Earth equals _____ hours.
4. The Earth's axis is _____, which affects the seasons.
5. The Sun appears to rise in the _____ and set in the _____.
6. The North Pole is at the _____ end of The Earth's axis.
7. Pakistan experiences its hottest temperatures in _____.
8. In winter, the Sun appears _____ in the sky.
9. The _____ Moon is when we cannot see the Moon at all.
10. The Moon takes about _____ days to complete one orbit around The Earth.
11. The _____ cycle is used in the Islamic calendar.
12. In _____, the Sun appears higher in the sky, resulting in longer days.



Tick (✓) for True and cross (x) for False statements.

1. In summer, the Sun appears lower in the sky. ()
2. Shadows are longest at midday. ()
3. The Full Moon is completely visible in the night sky. ()
4. The Earth's revolution around the Sun causes the change in seasons. ()
5. The Moon's shape does not change throughout the month. ()
6. Pakistan experiences snowfall in the northern areas during winter. ()
7. The Earth's axis is straight up and down. ()
8. The lunar cycle is important for marking Islamic months. ()



Multiple Choice Question.

1. What causes day and night on the Earth?

- a) Earth's revolution
- b) Earth's rotation
- c) The Moon's orbit
- d) The Sun's movement

2. Which phase of the Moon is completely invisible?

- a) Full Moon
- b) New Moon
- c) First Quarter
- d) Crescent Moon

3. The Earth takes _____ to complete one orbit around the Sun.

- a) 24 hours
- b) 30 days
- c) 365 days
- d) 12 hours

4. In which season does Pakistan experience the hottest temperatures?

- a) Winter
- b) Spring
- c) Summer
- d) Autumn

5. The Sun seems to move from _____ to _____ in the sky.

- a) North to South
- b) East to West
- c) South to North
- d) West to East



6. Which cycle determines the start of Islamic months?

- a) Solar cycle
- b) Lunar cycle
- c) Earth's rotation
- d) Seasonal cycle



Phases of the Moon

Write the phases of the Moon.





Activities



ACTIVITY

1.Shadow Tracking: Throughout the day, mark the length and direction of your shadow in the schoolyard. Observe how it changes from morning to evening.

ACTIVITY

2.Moon Observation: Over a month, observe and record the Moon's shape each night. Draw the phases and note when the new Moon appears.

